

Mike Mabey, Ph.D.

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Senior Python backend engineer with a Ph.D. in CS and 10+ years building high-throughput data pipelines, event-driven services, and internal developer tools. Open-source contributor (pyredox). Proven track record of 20×+ performance improvements and measurable cost savings across fintech, healthtech, and government.

EXPERIENCE

Backend Software Engineer

Macabacus Inc.

Apr 2025 – May 2026
(remote) Salt Lake City, UT

- Modernized 279 of 420 Python files (66% of the codebase) across 10 PRs — enforcing strict type annotations and **Ruff/Ty** linting rules, patching security vulnerabilities surfaced by automated review (including a latent SSRF risk in the Okta OAuth flow), and eliminating dozens of crashable runtime paths.
- Identified an async incompatibility in the planned **Celery** integration, pivoted to **Taskiq** after a successful POC, and led full implementation — including building progress monitoring and error handling not provided by the library — establishing patterns adopted by the rest of the team for subsequent background task work.

Senior Software Engineer, Back End

Capital One

Jan 2024 – Mar 2025
New York City, NY / (remote) Salt Lake City, UT

- Built a Python CLI tool that auto-exported **Pydantic** model hierarchies to the proprietary schema format used by the internal API registry — replacing a tedious, error-prone manual UI workflow and guaranteeing schema definitions stayed in sync with code.
- Refactored a **streamz** data pipeline from single-tenant to multi-tenant. Afterward, onboarded a client with more than five times any previous traffic volume in an effort projected to save \$2+ million annually.
- Discovered a bottleneck that could not scale horizontally, then purposefully applied Python's **asyncio** library and **SQS**'s batched submission to increase max throughput from 335 to >7,000 transactions per second.
- Designed and shipped 6–8 internal REST APIs on Capital One's data exchange platform, adapting to the company's proprietary systems, conventions, and pipeline constraints across varied use cases.
- Regularly mentored teammates on idiomatic Python — covering async patterns, type annotations, and tooling such as **Jinja** templates with **Pydantic** models — elevating the team's overall code quality.

Backend Engineer 3 - Product

Cedar Cares, Inc.

Jul 2019 – Jul 2023
Salt Lake City, UT

- Served as technical lead for integrations with Cerner and Epic — representing Cedar's capabilities to provider clients, negotiating API contracts with EHR vendors, and owning the compliance and certification submissions required to access regulated patient data under each vendor's authorization program.
- Eliminated manual JSON payload validation for the engineering team by conceiving, authoring, and open-sourcing **pyredox** — a Python library for generating Redox-spec-compliant payloads. Adopted by dozens of Cedar engineers; because Redox abstracts access to multiple EHR systems, the library reduced incremental onboarding effort for each additional Redox-based customer to near zero.
- Reduced client data refresh times from 30 minutes to seconds by applying **Domain-Driven Design** (DDD) patterns to refactor a batch-based Python service into an event-driven architecture.
- Built insurance claim adjudication tooling requiring deep ETL/ELT work across two complex data exchange standards — legacy **X12 EDI** and modern **FHIR** — using **Python**, **gRPC/protobufs**, **AWS (EC2, SQS, S3, and KMS)**, and **OpenCV**. Co-led a full rewrite of the data access layer to migrate to **Postgres**, enabling JSON-native storage for upcoming product requirements.

Computer Scientist (U.S. Army Civilian)

Data Science Directorate, Network Enterprise Technology Command (NETCOM)

Dec 2017 – Jun 2019

Phoenix, AZ

- Built the command's first-ever visibility into global help-desk performance — a dashboard visualizing SLA compliance across Army bases worldwide (thousands of tickets per base monthly, spanning all four military services) using **Python**, **Flask**, **Vue.js**, **pandas**, **Plotly**, **Celery**, and **Redis** — giving NETCOM leadership quantifiable insight into their primary mission for the first time.
- Pioneered DevOps practices within the Directorate — introducing **Git**, **GitLab**, and CI/CD pipelines to a geographically distributed team of 9 developers, reducing future deployments of big data analytic apps from a multi-week manual process to an automated, repeatable pipeline.
- Conducted the technical performance analysis for Sandia National Laboratories' annual contract review — converting anecdotal concerns into a documented, defensible case for non-renewal and saving the command \$3M annually.

EDUCATION**Ph.D. Computer Science — Information Assurance**

Arizona State University

Dec 2017

Tempe, AZ

M.S. Computer Science — Information Assurance

Arizona State University

Aug 2011

Tempe, AZ

B.S. Computer Science — Information Systems

Utah State University

May 2009

Logan, UT

TECHNICAL STRENGTHS

Languages	Python, SQL, Bash, HTML/CSS
Frameworks & Libraries	FastAPI, Flask, Pydantic, Jinja, streamz, pandas, Plotly
Async & Messaging	asyncio, Celery, Taskiq, SQS, RabbitMQ, AMQP, Redis
Cloud & Infrastructure	AWS (EC2, SQS, SNS, S3, KMS, DynamoDB), Azure, Docker
Databases	PostgreSQL, MySQL/MariaDB, SQLite
APIs & Data Formats	REST, gRPC/protobufs, FHIR, X12 EDI, JSON, XML
Testing & CI/CD	pytest, unittest, tox, nox, coverage.py, GitHub Actions, Docker, pre-commit
Dev Tools	Git, Ruff, Uv, Ty, Swagger, OpenCV